

Methods

Table 1: Surrogate hierarchy and data provenance. Each layer consumes data only from the layer above; no information flows upward or sideways. The neural network (planning model) never observes the RC plant’s realised outputs during closed-loop evaluation.

Model layer	Purpose	Data consumed	Role in training	Role in evaluation
EnergyPlus	Physical source	Weather files	None (upstream only)	None
Identified RC	Digital twin	Processed CSV	Data generator	Closed-loop plant
Neural network	Planning model	RC-generated data	Trained model	NMPC predictor

Table 2: Parameter perturbation ranges for the external-validity stress test. Each parameter is perturbed in its comfort-adverse direction for the one-at-a-time analysis; all four are perturbed simultaneously for the random ensemble and combined severe variants.

Parameter	Range	Adverse direction	Physical justification
C_{env}	$\pm 20\%$	+20% (slower response)	Longest time constant; least identifiable
R_{ex}	−15% to +10%	−15% (leakier building)	Infiltration varies seasonally
A_{sol}	± 0.10 (abs.)	−0.10 (less solar to air)	Solar gains hard to predict
K_{rad}	$\pm 10\%$	−10% (weaker radiator)	Fouling and air locks degrade output

Results

Discussion

The present study is a controlled digital-twin experiment. The perturbed-plant stress test confirms that the RAMC comfort advantage persists across all 14 plant variants tested, including 8 randomly sampled parameter vectors (mean $\Delta\text{CDH} = -3.50$ deg-h, all bootstrap CIs below zero). Under combined worst-case parametric mismatch, the advantage diminishes to -1.49 deg-h with a confidence interval crossing zero, indicating graceful degradation rather than reversal. The heat-delivery parameters (R_{ex} , K_{rad}) are the primary sensitivity drivers, as they directly affect the controller’s ability to translate heating actions into comfort outcomes. Validation against the full EnergyPlus simulator or physical hardware—where structural model mismatch is also present—remains a necessary next step.

Table 3: Perturbed-plant stress test results: paired differences (RAMC $\lambda=1.5\times 10^{-3}$ minus Fidelity Baseline) on the forecast-error scenario, 3 seeds per plant variant. Negative ΔCDH indicates RAMC advantage. Bold: 95% bootstrap CI excludes zero.

Plant variant	Category	ΔCDH (deg-h)	95% CI	ΔEnergy (kWh)
Matched	Baseline	-3.55	[-4.05, -3.27]	+1098
C_{env} +20%	One-at-a-time	-3.50	[-4.43, -3.01]	+1107
R_{ex} -15%	One-at-a-time	-2.13	[-2.89, -0.91]	+782
A_{sol} -0.10	One-at-a-time	-3.61	[-4.16, -3.24]	+1113
K_{rad} -10%	One-at-a-time	-1.93	[-3.33, +0.53]	+753
Random 01	Ensemble	-1.96	[-2.98, -1.45]	+1121
Random 02	Ensemble	-5.62	[-6.55, -4.66]	+1120
Random 03	Ensemble	-3.08	[-3.48, -2.84]	+1251
Random 04	Ensemble	-4.84	[-6.17, -2.61]	+936
Random 05	Ensemble	-4.06	[-4.42, -3.86]	+1053
Random 06	Ensemble	-3.28	[-6.46, -1.53]	+843
Random 07	Ensemble	-2.65	[-3.55, -2.08]	+1129
Random 08	Ensemble	-2.51	[-3.78, -0.98]	+796
Combined severe	Worst case	-1.49	[-3.66, +1.19]	+269
<i>Random ensemble aggregate</i>		-3.50	<i>all 8/8 CIs < 0</i>	+1031

Table 4: Summary of the external-validity stress test. The RAMC advantage persists across all 14 plant variants (14/14 negative mean ΔCDH , 12/14 with CI excluding zero).

Category	Variants	Mean ΔCDH	CI < 0	Mean ΔEnergy
Matched	1	-3.55	1/1	+1098 kWh
One-at-a-time	4	-2.79	3/4	+939 kWh
Random ensemble	8	-3.50	8/8	+1031 kWh
Combined severe	1	-1.49	0/1	+269 kWh
Total	14	-3.12	12/14	+960 kWh